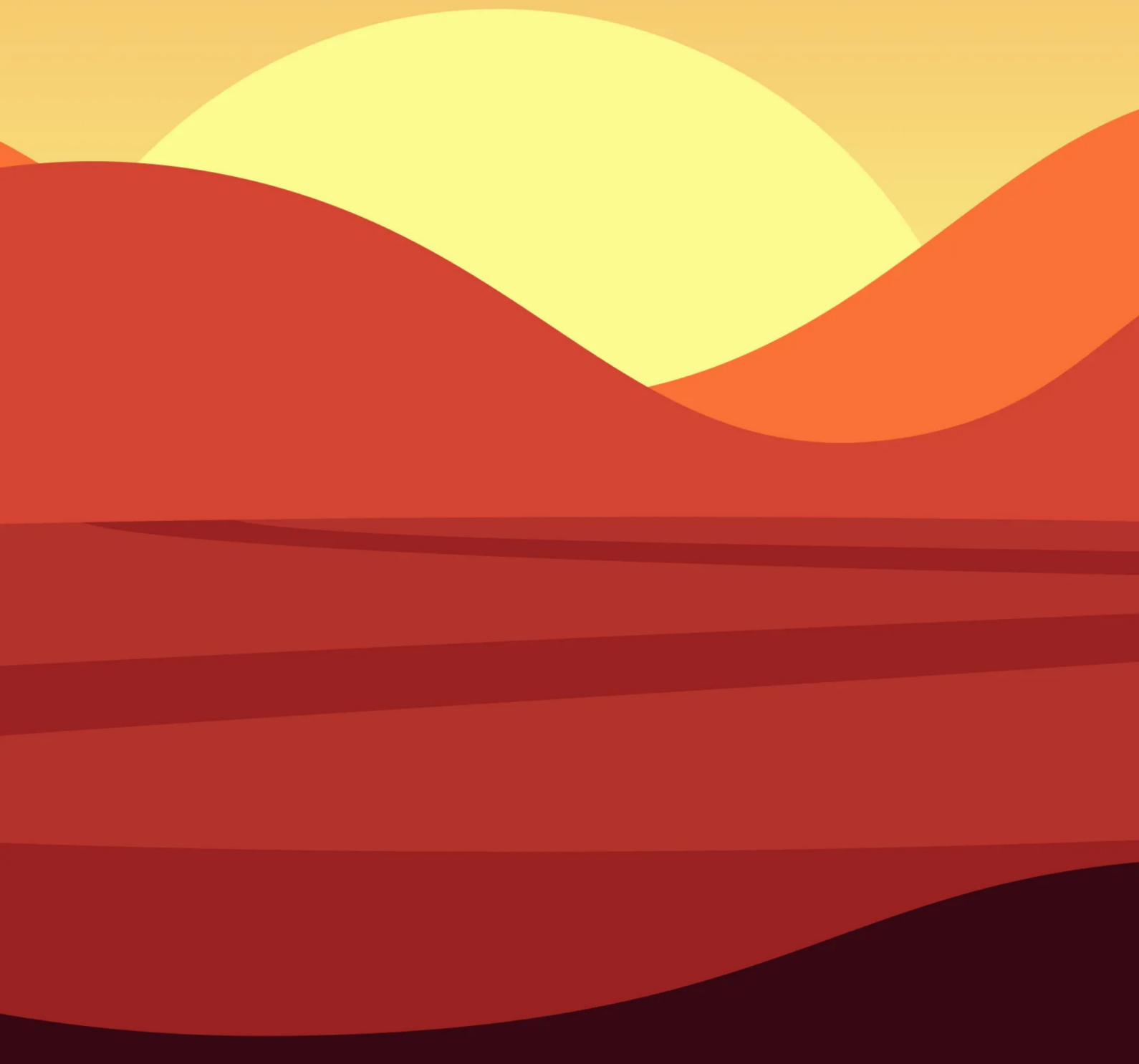


Title:



# *Automata Theory*



13/08/26

### Pumping Lemma:

For any regular language  $L$ , there is a magic number ' $p$ ' the pumping length. Any string ' $s$ ' in  $L$  with a length of  $p$  or more " $|s| \geq p$ " can be split into 3 parts  $s = xyz$

Conditions for splitting

$$\textcircled{1} xy^iz \in L \quad \forall i \geq 0$$

$$\textcircled{2} |y| > 0$$

$$\textcircled{3} |xy| \leq p$$

pumping length is minimum length needed to put a string through machine

Pumping lemma can only be used to prove a grammar is not regular

Q) Check whether Language  $L = \{a^n \mid n \geq 0\}$  is a regular language using pumping lemma

Sol<sup>n</sup>: Let ' $p$ ' be pumping length

Choosing a string  $s = a^p$

$$\Rightarrow |s| = |a^p| = p \geq p$$

i) Split  $s$

$$|xy| \leq p \quad |y| = k \geq 1$$

$$\text{Let } |x| = a, |y| = k \text{ \& } |z| = |s| - a - k$$

$$\Rightarrow |z| = p - a - k$$

$s$  has only  $a$ 's so there are no different cases.

ii) wkt  $1 \leq k \leq p$  (since  $|xy| \leq p$  \&  $|y| = k \Rightarrow |y| \leq |xy| \Rightarrow k \leq p$ )

iii) For  $i=2$  in  $xy^iz$

$$= xy^2z$$

$$\Rightarrow |xy^2z| = p! + k \quad (|xyz| = p! \text{ \& } |y| = k)$$

iv) Finally, we need to prove  $p! + k \neq (\text{natural number})!$

$$\underbrace{p! < p! + k \leq p! + p < (p+1)!}_{\text{obvious } k \leq p \quad \text{Math}}$$

$$\Rightarrow p! < p! + k < (p+1)!$$

$$p! + k \neq p! \text{ or } (p+1)!$$

$$\therefore xy^2z \notin L$$

$\therefore L = \{a^n \mid n \geq 0\}$  is not a regular language